IrisPD v14

Windows pupil-distance measurement and experimental eye-alignment monitoring

HELP & REFERENCE • NOVEMBER 2026

Read this first

IrisPD uses a webcam and local MediaPipe iris landmarks to estimate pupillary distance (PD) and to observe changes in relative eye position. **It is not a validated ophthalmic device and cannot diagnose intermittent strabismus.** The eye-alignment display is an experimental movement indicator, not a clinical angle or prism-diopter measurement.

Quick start

• **Install:** unzip the full package to a stable Windows folder; launch **Run_IrisPD_Windows.bat** with 64-bit Python 3.11. The included EXE builder can produce a standalone executable on a Windows PC.

• **One-time PD:** choose Camera index, click Start webcam, look at the lens and wait for the configurable frame median. Alternatively use Open photo.

• **Continuous PD:** choose Limits / alerts / CSV, set the range, interval, frames and alerts, then click Start monitoring.

• **Eye alignment:** click Start alignment, look at one stationary fixation target and let the initial reference complete. Recalibrate if desired.

• **Hardware control:** Pause camera physically releases webcam access so another application can use it; Resume camera reopens it.

• **Layout:** the second Reorient preset (monitors on top, camera left, PD monitor left) is the v12 default. All eight choices remain available; F11 enters full display.

Privacy: Camera imagery is processed locally and not automatically saved or uploaded. CSV is saved only when logging is enabled. A configured Buy Me a Coffee link opens a browser only when clicked.

1 Camera, measurements and scale method

Camera preview	Mirrored live image with iris marks and estimated PD. Hide preview reduces rendering overhead but does not release the hardware.
Start webcam	Opens the selected camera. A one-off capture uses 1-500 accepted frames (30 by default) and locks the median.
Open photo	Loads one local photo; no upload. Use a frontal, evenly lit face.
Freeze for card	Freezes the currently displayed webcam frame. A real card must be visible at eye depth. Live capture stops while frozen.
Mark card edges	On a frozen frame or photo, click the two endpoints of the SAME 85.60 mm long card edge.
Reset / measure again	Clears the held measurement; it also stops continuous PD monitoring if that monitor is active.
Save result CSV	Exports a one-off measurement to a new file; refuses to overwrite an existing file.
Camera index	Choose 0, 1, 2 or 3, then reopen the camera to use the chosen device.
Capture progress	Displays collection progress toward the requested rolling median, or why frames are being excluded.
Alignment / Quality	Fixed-size scrollable warning box. Longer notes do not move or resize the layout.

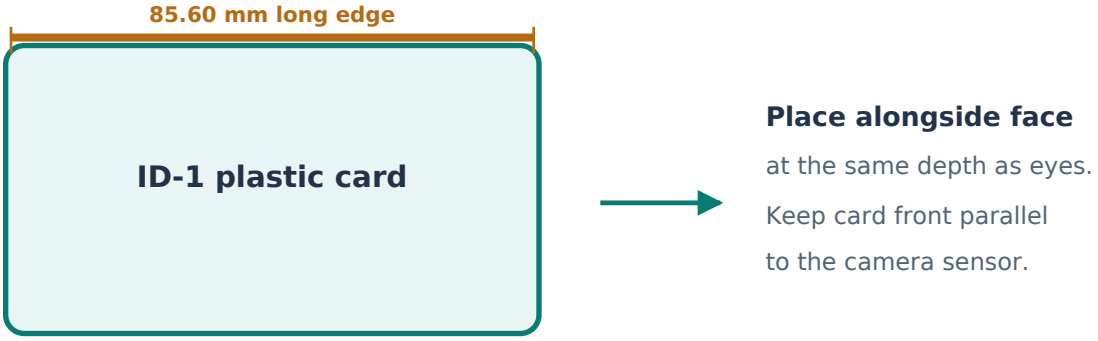
Understanding the measurement method (removed from the main GUI in v12)

Iris-based scale: Webcam PD is estimated from pupil/iris-center spacing in pixels, using an assumed adult iris diameter of about **11.7 mm**. Real iris size varies, so live PD is approximate. **Card-calibrated still image:** a known 85.60 mm card edge sets pixels per millimeter for that frozen frame or photo only. The live continuous-monitoring scale is not permanently recalibrated by that still-image operation.

The measurement **Method** is intentionally absent from the compact PD panel. The method remains a column in exported CSV files; see this description to interpret iris-scale versus card-calibrated results.

2 Calibrating a frozen picture with a card

Use a standard ISO ID-1 size card with a long edge of **85.60 mm** (for example, a blank plastic ID-size card). Avoid exposing a payment card number or other sensitive information to the camera, even though the program does not upload images.



• **1 — Prepare:** Select Start webcam and position your face 40–60 cm from the lens. Use even light and keep the head facing forward.

• **2 — Position the card:** Hold the entire plastic card next to your cheek, in roughly the same plane/depth as your eyes, with its flat front facing the lens. Do not tilt the card toward or away from the camera. Keep both eyes visible.

• **3 — Freeze:** While holding the position, select **Freeze for card**. You can alternatively select Open photo if your face and card are visible together in a saved image.

• **4 — Mark:** Click Mark card edges if necessary. Click **one corner** at the beginning of the card's 85.60 mm edge, then click **the other corner** at the far end of that SAME long edge. Do not click its diagonal or short edge.

• **5 — Verify:** Read the new calibrated PD. To correct a misplaced point, choose Mark card edges and repeat. Select Save result CSV for the still-image measurement.

• **6 — Resume:** Use Start webcam to return to live iris-scale estimates; the card scaling is not transferred to the rolling webcam monitor.

Accuracy cautions: At different depths, pixels per millimeter differ; a card nearer the camera makes PD appear wrong. A tilted card makes its projected edge too short. The method is a convenience estimate, not a substitute for an optician's optical PD measurement.

3 Compact PD panel and continuous monitor

Screen-left / right	Smaller monocular estimates at either side of the smaller total binocular PD. These estimates are based on a nose-bridge approximation and are not anatomical or dispensing-grade labels.
Total PD	Center number in millimeters; use frontal, steady fixation. Values below or above limits are heuristic screening thresholds, not strabismus evidence.
Start / Stop monitoring	Schedules checks continuously until stopped, paused, or camera is released. The eye-alignment monitor can run simultaneously.
Alert BELOW / ABOVE	Lower/upper binocular PD limits; defaults 60 mm and 70 mm. Values exactly on the limit are not out of range.
Check every (seconds)	Sampling interval; default 3 seconds. Each check uses the most recent accepted rolling frames.
Consecutive checks	Required repeated valid readings on the same side of the limit; default 3. A missing reading breaks confirmation.
Frames per reading	1–500, default 30. A single frame cannot establish temporal stability.
Ignore poor alignment	Optional bypass for some geometry warnings. Does NOT bypass hard missing-face / both-eyes-looking-away / strong-head-motion rejection.
Ignore unstable median	Optional bypass for rolling jitter. Results can be less trustworthy and retain a warning.

Alarms and CSV

Choose any combination of colored visual alert, Windows sound, draggable popup, and Windows spoken alert. Confirmed PD and eye-alignment popups close automatically after **4 seconds**. Repeat notifications may be enabled. With automatic CSV enabled, the application appends timestamp, PD, sides, scale method, threshold state, validation status and quality notes to the configured file; earlier rows are retained.

No reading is not an alarm. When one or both irises are missing, the face turns away, the eyes blink or both gaze away, or the motion gate is active, no PD threshold alert fires. Such checks may be logged as NO_READING. The checks do not prove the eyes are correctly aligned.

4 Experimental eye-alignment monitor

The eye monitor analyzes each pupil/iris independently relative to its own eye corners and reports normalized differential position in **percent of eye width**. It is not a clinical measurement of gaze angle, deviation in prism diopters, or evidence that a particular eye is responsible.

Start alignment	Begins tracking. A new application session requires a reference; first calibration normally collects 45 good fixation frames (user-set 10–300).
Recalibrate	Manually replaces the reference while you keep both eyes fixed on one stationary target near the camera.
Stop / Settings	Stop this monitor independently of PD, or edit reference, movement and alert settings after stopping monitoring.
Difference threshold	Default 12% eye width (range 3–45%); candidate movements are not yet confirmed events.
Confirmation / recovery	Default 0.65 s sustained difference and at least 3 frames; recovery default 0.60 s.
Head-position tolerance	Relative yaw/roll tolerance (0.015–0.2); a head-pose mismatch normally pauses detection rather than implying eye drift.
Auto recalibrate (v12)	**OFF by default.** If checked, stable displacement relative to the reference must persist for the selected 1–15 s (2.5 s default), then one new reference is collected. An additional 12-second trigger cooldown prevents immediate repeats.
Changing head pose	With auto recalibration OFF, the original reference is preserved. With it ON, transient movement still only pauses; prolonged stable reorientation may trigger a new reference.
Event and sample CSV	Append-only EyeAlignment_events.csv and optional EyeAlignment_samples.csv, separate from PD measurements. Configure CSV folder and sample interval 0.2–60 s.
Alert types / Auto-start	Set visual/sound/popup/voice alerts and optional alignment monitoring at application startup independently of PD settings.

Best practice: Calibrate only when looking steadily at a single stationary fixation target with the eyes in their usual alignment. Auto-reorientation can hide a real eye-position change if calibration occurs during an episode; turn it OFF if you want a consistent baseline across time.

5 Camera pause, cooling and low-power operation

Pause camera	Top-right control: stops both monitors and immediately calls VideoCapture.release(), discarding the active frame. The camera is freed for other applications (subject to Windows driver behavior). No monitoring alerts occur while paused.
Resume camera	Reopens the camera and restores the monitor(s) previously active. Other applications might need to release the device before reopening succeeds.
Pause for…	User-selected 0.01–168 hours. Release is immediate; timed resume occurs if IrisPD is running, or when it next opens after the deadline.
Hide / Show preview	Stops or resumes visible webcam rendering. Tracking continues and the webcam remains occupied: use Pause camera if another application needs it.
Camera power	Configures duty cycling: default 30 seconds ON and 15 seconds released/OFF during active monitoring. Toggle OFF for uninterrupted observation.
Camera ON/OFF times	Choose 10–3600 s ON and 5–3600 s OFF. The webcam is physically released in the OFF segment, not merely hidden.
Head-motion settling	Selectable 0.3–5.0-second wait after substantial head movement (default 1 s), to avoid collecting invalid frames or eye events.
Cooling/calibration	Cooling preserves an existing alignment reference and interrupts candidate event timers. It does NOT automatically recalibrate solely because the webcam reopened.
Camera index	Choose 0–3 and restart camera access to switch the hardware.

Cooling periods can miss brief intermittent eye-wandering events. The CPU-saving inference pipeline requests 960 × 540 at a target 10 FPS, limits FaceMesh processing to 640 px width and slows preview rendering. Real hardware use and temperature vary by camera.

6 Reorientation, full display and startup

Reorient	Choose the **top row**, **camera side**, and **eye-alignment side** independently, or select one of eight presets. **Preset #2** (monitors top, camera left, PD monitor left) is the default for new and upgraded v12 installations. Your subsequent selections are saved.
Stable panel sizes	The warning box has a fixed height, and text changes cannot resize panel splits. Panels adapt only to deliberate window resizing and changing orientation.
F11 / Escape	F11 enters full display; Esc exits. Windows maximize is also supported.
FHD / 2.8K / 4K	The panel splitter adapts to approximately 1920×1080, 2880×1800 and 3840×2160 displays, accounting for DPI/Windows scaling.
Launch on Windows sign-in	Per-user Windows startup option in the PD settings. Keep the extracted folder at a stable location; re-save the setting if you move it.
Auto-start PD / alignment	Each monitor has its own saved auto-start switch; both may start from the same webcam at application launch.
Help PDF	Header button opens this local manual in your default PDF viewer.

Buy Me a Coffee — optional 60-second support link

The support controls appear **midway through both settings dialogs**, under Limits / alerts / CSV and Eye alignment Settings. A shared support button lives **in the header between the IrisPD logo and menu controls**. It counts down from 60 seconds and then disappears without moving other panels. The banner is optional; toggle it OFF in either settings dialog.

The support button opens https://buymeacoffee.com/nipu011 in your default browser. Show or hide the 60-second banner using either monitor settings window. This support feature does not send webcam images or collect payment information.

Standalone Windows EXE

To create one file that runs without a separate Python installation, run Build_Windows_Standalone_EXE.bat on Windows with Python 3.11 and internet access. The output is dist\IrisPD.exe. Alternatively, upload the source to GitHub and run the bundled Windows build workflow. See STANDALONE_WINDOWS_EXE.md.

7 Troubleshooting, data and medical limitations

No face / poor iris detail	Improve room light; remove glare; face the lens; keep both eyes visible; choose an appropriate camera index.
PD unexpectedly changes	Check head motion, camera distance, eyelids, iris landmark stability and bypass options. One-off card calibration is not persistent across video frames.
Too many eye calibration messages	Leave auto recalibration OFF unless you explicitly want new baselines after stable head reorientation. Keep the same fixation target.
Monitoring pauses after a head turn	This is intended behavior for image-based estimates. Return to the reference pose or use manual Recalibrate; optional auto-reorientation changes the baseline after a sustained pause.
Windows camera in use	Click Pause camera (NOT Hide preview). The program calls release and stops its camera loop. Another application may need a moment to detect the freed device.
Camera cannot reopen	Close the other app using the device, verify Windows camera permission, then click Resume camera or Start webcam.
No CSV rows	Check that logging is enabled and that the configured folder is writable. Existing files with incompatible column headers are rejected rather than overwritten.
Support link does not open	The button always opens https://buymeacoffee.com/nipu011. If it fails, check the default web browser settings.

Clinical note

Binocular pupillary distance is mainly a geometric fitting measurement. It is **not a reliable strabismus screen**. Eye-alignment tracking is also experimental: head pose, gaze changes, blinks and imperfect landmarks can mimic eye drift. If you have intermittent eye wandering, a specialist eye examination (including cover testing) is more informative. New sudden double vision or eye deviation, particularly with neurological symptoms, requires prompt medical evaluation.